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Research papers

Multi-objective optimization of start-up strategies for pumped storage units with hybrid variable and fixed-speed configuration

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ABSTRACT

Variable-speed pumped storage technology demonstrates superior advantages in rapid response, operational flexibility, reliability, and efficiency, representing the new developmental direction in pumped storage technology. However, current research lacks investigation on start-up strategies for hybrid variable/fixed-speed units in one-penstock-two-unit systems. To address the optimal start-up strategy selection for such hybrid configurations, this study proposes a multi-objective start-up optimization strategy based on mathematical models of both variable-speed and fixed-speed units. First, considering the nonlinearity of the water conveyance system and pump-turbine dynamics, a refined hydro-mechanical-electrical coupling model is established to simulate the operational states of both variable-speed and fixed-speed units. Second, the variable-speed start-up strategy incorporates speed-related parameters. Building on this foundation, six start-up strategies are evaluated by optimizing the speed ramp-up time and integral of time-weighted absolute error (ITAE) index, with parameter selection focusing on guide vane opening, controller closed-loop engagement point, and speed setpoint for variable-speed units. A combined AHP-entropy weight-game theory method is introduced for optimal strategy selection. The results indicate that adopting non-rated speed setpoints in variable-speed unit start-up strategies effectively prevents operation in the reverse "S" characteristic region. Among the six strategies, different parameter configurations of the two-stage guide vane control law prove optimal, significantly improving dynamic response quality during the start-up process of heterogeneous units sharing the same penstock. These optimization outcomes provide decision-making support for formulating start-up control strategies in hybrid variable/fixed-speed configurations.

1. Introduction

Driven by the "Dual Carbon" goals, constructing a new power system dominated by green renewable energy has become the core direction of China's energy revolution [1]. However, challenges such as low comprehensive utilization efficiency and poor coordination of renewable energy sources have led to increasing peak-shaving and frequency-regulation pressures on the grid, weakening system inertia support capabilities and gradually degrading stability [2]. As an energy storage technology, pumped storage power stations represent an inevitable product of the interactive development between grid power supply structures and load demand [3], playing an indispensable role in ensuring the secure and stable operation of new power systems with high renewable energy penetration [4,5]. Notably, to promptly respond to dynamic load variations in modern power grids, pumped storage units (PSUs) frequently undergo operational mode transitions, with start-up operations occurring particularly often during scenarios of drastic renewable power fluctuations. Consequently, the safe and stable start-stop operation of PSUs is critically significant for enhancing the complementary efficiency between renewable energy and modern power grids, as well as strengthening the grid's capacity to accommodate intermittent energy [6].

The novel variable-speed pumped storage units (VS-PSUs), with their superior efficiency, rapid response, operational flexibility, and reliability, are not constrained by fixed-speed operation, representing a pivotal development direction and major research focus in global

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pumped storage technology [7]. Capable of maintaining optimal rotational speeds and peak efficiency across varying head conditions and generation modes, VS-PSUs exhibit significant potential for enhancing the stability of modern power systems. Existing research has investigated VS-PSU start-up operations: Zhao et al. [8] conducted model experiments at different heads and speeds, analyzing no-load characteristics to reveal the intrinsic evolution patterns of variable-speed operation. They proposed quantitative metrics (migration distance and operating-point slope) for no-load performance evaluation and developed a rapid generation start-up strategy (SUS) integrated into control systems, providing valuable references for start-up optimization. Feng et al. [9] analyzed the operational zones of VS-PSUs in turbine mode, elucidating mechanisms to avoid the S-characteristic region while quantitatively comparing control system performance with conventional units. Hu et al. [10] demonstrated that 300 MW VS-PSUs achieved fast grid voltage synchronization using grid-voltage-oriented vector control during no-load start-up. Pannatier et al. [11] and Shao et al. [12] respectively investigated pump-mode start-up synchronization and transient behaviors. However, these studies predominantly assumed simplified single-unit-per-penstock configurations. In practical engineering, the one-penstock-two-unit layout prevails, where shared water conveyance pipelines inevitably cause hydraulic interference during operation—particularly during large transient start-up processes. For hybrid variable/fixed-speed units in such configurations, the mutual hydraulic influences during start-up demand sophisticated control strategies and parameter designs to ensure rapid, stable, and efficient coordinated operation of both units.

Previous research on start-up strategies and optimization for PSUs has predominantly employed single-unit-per-penstock models, typically adopting either open-loop or closed-loop start-up control methods. Liu et al. [13] developed a model-free adaptive controller with parameter optimization, comparing its performance with PID, fractional-order PID, and nonlinear generalized predictive control across four start-up scenarios. Their results demonstrated significant improvements in both speed and pressure control. Xu et al. [14] proposed a two-stage SUS combining fuzzy fractional-order PID control with multi-objective grey wolf optimization, which enhanced dynamic stability and control performance during low-head start-up compared to single-stage methods. Wang's team implemented an artificial sheep flock optimization-based two-stage closed-loop strategy that substantially reduced speed rise time and overshoot, achieving superior start-up transients. Xu et al. [5] established an instantaneous linearized ARIMA model for PSUs and designed a novel adaptive conditional predictive fuzzy PID controller optimized through bacterial foraging algorithm. For one-penstock-two-unit systems, Hou et al. [15] investigated sequential start-up timing effects and developed a multi-objective particle swarm optimization strategy for single-stage sequential start-up control, improving transition quality. Li et al. [16] applied multi-objective artificial sheep flock optimization to sequential start-up under low-head conditions, optimizing both speed rise time and integral of time-weighted absolute error (ITAE). Feng et al. [17] proposed a refined modeling-based multi-objective optimization strategy for back-to-back start-up at low heads, evaluating four control laws through entropy-weighted ranking, effectively preventing operation in the reverse S-characteristic region. Notably, these studies universally employed fixed-speed unit models without considering the impact of variable-speed characteristics on start-up processes. This limitation becomes particularly significant in hybrid variable/fixed-speed configurations where the interaction between different operational paradigms necessitates specialized control approaches. The absence of variable-speed dynamics in existing models overlooks critical transient behaviors during coordinated start-up scenarios in complex hydraulic systems.

To address these research gaps, this study establishes a full-size converter-based hydro-mechanical-electrical coupled time-varying nonlinear model for variable-speed units. The hydraulic interference between variable-speed and fixed-speed units during co-start-up in hybrid configurations significantly complicates operational control. Accordingly, we propose a novel start-up optimization strategy for hybrid-configured PSUs, implementing multi-objective optimization across six typical start-up schemes and selecting the optimal strategy through combined analytic hierarchy process-entropy weight-game theory evaluation. This approach demonstrates substantial engineering applicability by providing technical support for developing coordinated start-up control strategies in shared-penstock systems containing variable-speed units. The key innovations of this research are summarized as follows:

(1) Development of a full-size-converter-based hydro-mechanical-electrical coupled time-varying nonlinear model.
(2) Proposal of a start-up optimization strategy for hybrid variable/fixed-speed pumped storage configurations.
(3) Incorporation of variable-speed characteristics into SUS optimization control.
(4) Establishment of a start-up optimization framework for full-size converter variable-speed units (FSC-VSUs), including a decision-making model for selecting optimal non-dominated start-up solutions.

The structure of this paper is organized as follows: Section 2 establishes a refined mathematical model for the start-up process of FSC-VSUs; Section 3 proposes a multi-constrained start-up optimization strategy for hybrid variable/fixed-speed PSUs, which innovatively develops a start-up decision-making model for optimal strategy selection; Section 4 validates the effectiveness of the start-up optimization framework through comparative simulation results; and Section 5 concludes the research findings.

2. Refined modeling of start-up process for FSC-VSPSUs

As a time-varying, nonlinear, and complex hydro-mechanical-electrical coupled system, a pumped storage power station primarily consists of a water conveyance system, pump-turbine, governor, generator/motor, and excitation system [18]. In plants incorporating full-size converter units, the configuration additionally requires back-to-back full-size converters compared to conventional units. Fig. 1 illustrates the unique one-penstock-two-unit heterogeneous configuration, which comprises: an upstream reservoir, upstream surge chamber, pressure conduit, two symmetrically arranged units (the fixed-speed unit containing a pump-turbine, generator, excitation system and governor; the variable-speed unit adding a full-size converter with control system), downstream surge chamber, draft tube pressure conduit, and downstream reservoir. To simulate the start-up transient processes in this heterogeneous system, this section develops refined models of: (1) the water conveyance system, (2) pump-turbine, (3) governor, (4) generator, (5) excitation system, and (6) machine-side converter control system, thereby establishing the modeling foundation for subsequent multi-objective optimization studies. It should be noted that the studied pumped storage power station is originally equipped with fixed-speed units. In this study, the model assumes a variable-speed transformation of the existing station to investigate the hydraulic dynamic responses under variable-speed operation, while the structural and hydraulic parameters remain identical to those of the

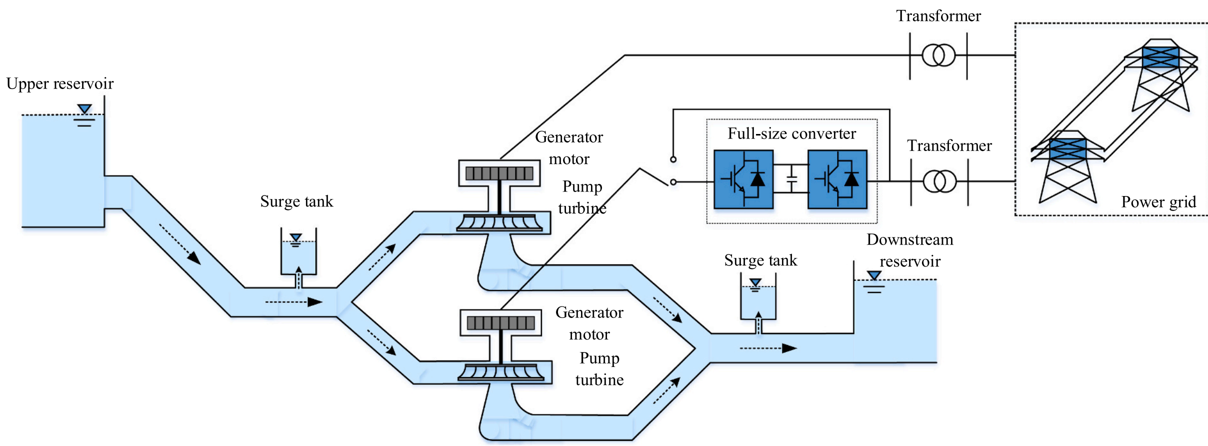

Fig. 1. Schematic diagram of the grid-connected arrangement of the same pipe and different machines with FSC-VSPSUs.

original fixed-speed configuration.

2.1. Diversion system model

The water conveyance system exhibits complex fluid dynamic behavior due to water inertia, elastic water column effects, and frictional resistance. During large transient start-up processes where system parameters undergo significant variations, the conventional elastic water hammer model fails to accurately capture hydraulic characteristics. Therefore, the method of characteristics [19] should be employed to establish the pressure conduit's mathematical model. The fundamental governing equations for one-dimensional unsteady flow in the conveyance system - the momentum equation and continuity equation - are expressed as follows [20]:

$$\begin{cases} \dfrac{\partial H(x,t)}{\partial x} + \dfrac{1}{gA}\dfrac{\partial Q(x,t)}{\partial t} + \dfrac{f|Q(x,t)|}{2gDA^2}Q(x,t) = 0 \\ \dfrac{a^2}{gA}\dfrac{\partial Q(x,t)}{\partial x} + \dfrac{\partial H(x,t)}{\partial t} = 0 \end{cases} \tag{1}$$

Where H represents the pressure head in the water conveyance system; Q denotes the flow rate through the pressure conduit; x is the axial coordinate along the pipeline length; t indicates time; A corresponds to the cross-sectional area of the conduit; a is the water hammer wave speed; D signifies the pipe diameter; g stands for gravitational acceleration; f represents the Darcy-Weisbach friction factor.

When solving Eq. (1) using the method of characteristics, the characteristic Equation are introduced, transforming the partial differential equations of Eq. (1) into:

$$\begin{cases} \dfrac{dH}{dt} \pm \dfrac{a}{gA}\dfrac{dQ}{dt} \pm \dfrac{af}{2gDA^2}|Q|Q = 0 \\ \dfrac{dx}{dt} = \pm a \end{cases} \tag{2}$$

In Eq. (2), $dx/dt = \pm a$ represents the propagation direction of the pressure wave along the characteristic lines C^+ and C^-, where a denotes the wave velocity in the conduit. This relation originates from the method of characteristics, showing that disturbances travel along these lines with the wave velocity.

Based on the finite difference method, a characteristic grid is constructed, and Eq. (2) is transformed into a difference equation. This allows for the calculation of the flow rate and head in the pipeline at any given time.

Pumped storage power plants with shared-penstock heterogeneous-unit configurations incorporate bifurcated pipes and converging pipes. The bifurcated pipes distribute water from the common headrace tunnel to individual units, while the converging pipes collect discharge from each unit into the shared tailrace tunnel. A schematic of these piping arrangements is presented in Fig. 2.

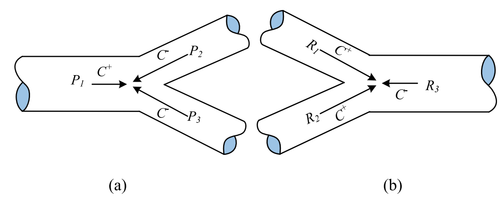

Fig. 2. Schematic diagram of bifurcated and converging pipelines: (a) bifurcated pipe; (b) converging pipe.

The characteristic equations for the bifurcated pipe and converging pipe are respectively expressed as:

$$\begin{cases} Q_{t+\Delta t}^{P_1} = C_m^{P_1} - C_{aP_1} H_{t+\Delta t}^{P_1} \\ Q_{t+\Delta t}^{P_2} = C_n^{P_2} + C_{aP_2} H_{t+\Delta t}^{P_2} \\ Q_{t+\Delta t}^{P_3} = C_n^{P_3} + C_{aP_3} H_{t+\Delta t}^{P_3} \end{cases} \tag{3}$$

$$\begin{cases} Q_{t+\Delta t}^{R_1} = C_N^{R_1} + C_{aR_1} H_{t+\Delta t}^{R_1} \\ Q_{t+\Delta t}^{R_2} = C_M^{R_2} - C_{aR_2} H_{t+\Delta t}^{R_2} \\ Q_{t+\Delta t}^{R_3} = C_M^{R_3} - C_{aR_3} H_{t+\Delta t}^{R_3} \end{cases} \tag{4}$$

In Eqs. (3) and (4), C_{aPi} and C_{aRi} ($i = 1, 2, 3$) denote the characteristic constants determined by the upstream and downstream boundary conditions of the bifurcated and converging pipes, respectively. These constants are derived through the method of characteristics, ensuring that the continuity and momentum equations are satisfied simultaneously under transient conditions.

The hydraulic pressure and flow rate equations at the common junction P of the bifurcated pipe are expressed as:

$$\begin{cases} Q_{t+\Delta t}^{P_1} = Q_{t+\Delta t}^{P_2} + Q_{t+\Delta t}^{P_3} \\ H_{t+\Delta t}^{P_1} = H_{t+\Delta t}^{P_2} = H_{t+\Delta t}^{P_3} = H_{t+\Delta t}^{P} \end{cases} \tag{5}$$

The hydraulic pressure and flow rate equations at the common junction R of the converging pipe are formulated as:

$$\begin{cases} Q_{t+\Delta t}^{R_1} + Q_{t+\Delta t}^{R_2} = Q_{t+\Delta t}^{R_3} \\ H_{t+\Delta t}^{R_1} = H_{t+\Delta t}^{R_2} = H_{t+\Delta t}^{R_3} = H_{t+\Delta t}^{R} \end{cases} \tag{6}$$

2.2. Pump-turbine model

To resolve the multivalued problem in the reverse S-characteristic region of the complete characteristic curves, this study processes the curves using an improved Suter transformation method, whose formulation is given by [21]:

$$\begin{cases} WH(x,y)=\dfrac{h\left(y+C_y\right)^2}{n^2+q^2+C_h h} \\ WM(x,y)=\dfrac{(m+k_1 h)\left(y+C_y\right)^2}{n^2+q^2+C_h h} \\ x=\arctan\left[\left(q+k_2\sqrt{h}\right)/n\right], n\geq 0 \\ x=\pi+\arctan\left[\left(q+k_2\sqrt{h}\right)/n\right], n<0 \end{cases} \tag{7}$$

where *WH(x, y)* and *WM(x, y)* are the variables describing the complete characteristic curves after the improved Suter transformation, and x is the independent variable; h, y, n, q, and m represent the relative values of head, guide vane opening, rotational speed, flow rate, and torque, respectively; $k_1 > |M_{11\max}|/M_{11r}$; $k_2 = 0.5 \sim 1.2$; $C_y = 0.1 \sim 0.3$; $C_h = 0.4 \sim 0.6$。The complete characteristic curves processed by the improved Suter transformation method are presented in Fig. 3.

2.3. Modeling of the governor system

The governor systems of PSUs predominantly employ PID control strategies, whose transfer function is expressed as:

$$u=\left(K_p+\frac{Ki}{s}+K_d s\right)\left(n_{ref}-n\right) \tag{8}$$

where K_p, K_i, and K_d represent the proportional, integral, and derivative coefficients of the parallel PID controller, respectively; s is the laplace transform operator, and u is the output signal of the PID controller.

The transfer function of the governor actuator is given by [22]:

$$\frac{\mathrm{d}y}{\mathrm{d}t}=\frac{1}{T_y}(u-y) \tag{9}$$

where T_y is the response time constant of the main servomotor.

2.4. Synchronous generator model

Synchronous generators, which convert mechanical energy into electrical energy, play a pivotal role in pumped storage power stations. To accurately characterize the dynamic behavior of FSC-VSUs during start-up transients, this study develops a synchronous generator mathematical model that comprehensively accounts for electromagnetic and rotor dynamics across the d, q, and f windings. The established model consists of three fundamental components: the flux linkage Eq. (10), voltage Eq. (11), and rotor motion Eq. (12) [10,23].

$$\begin{cases} u_d=\dfrac{\mathrm{d}\psi_d}{\mathrm{d}t}-r_s i_d-\omega\psi_q \\ u_q=\dfrac{\mathrm{d}\psi_q}{\mathrm{d}t}-r_s i_q+\omega\psi_d \\ u_f=\dfrac{\mathrm{d}\psi_f}{\mathrm{d}t}+r_f i_f \end{cases} \tag{10}$$

$$\begin{cases} \psi_d=-L_d i_d+m_{af} i_f \\ \psi_q=-L_q i_q \\ \psi_f=-\dfrac{3}{2}m_{af} i_d+L_f i_f \end{cases} \tag{11}$$

$$\begin{cases} \dfrac{\mathrm{d}\omega}{\mathrm{d}t}=\dfrac{M_t-\left(\psi_d i_q-\psi_q i_d\right)-D_f(\omega-1)}{T_a} \\ \dfrac{\mathrm{d}\delta}{\mathrm{d}t}=\omega_0(\omega-1) \end{cases} \tag{12}$$

Where u_d and u_q are the voltage components in the dq coordinate system; ψ_d and ψ_q denote the flux linkage components in the dq coordinate system; i_d and i_q are the current components in the dq coordinate system; r_s is the stator resistance; ω is the angular velocity; u_f is the excitation voltage; $\psi_f$is the excitation flux linkage; r_f is the excitation winding resistance; i_f is the excitation current; L_d and L_q are the inductance components in the dq coordinate system; m_{af} is the mutual inductance between the stator and excitation winding; L_f is the excitation winding inductance; M_t is the output torque of the pump-turbine; T_a is the inertia time constant of the unit; D_f is the damping coefficient of the synchronous generator; δ is the generator power angle; and ω_0 is the rated angular velocity.

2.5. Excitation system model

The mathematical model of the thyristor-based excitation regulator adopted in this study is derived from the IEEE model library, explicitly incorporating exciter saturation characteristics [17]. The schematic diagram of the excitation system model architecture is presented in Fig. 4.

Where T_b and T_c represent the respective time constants for the lead

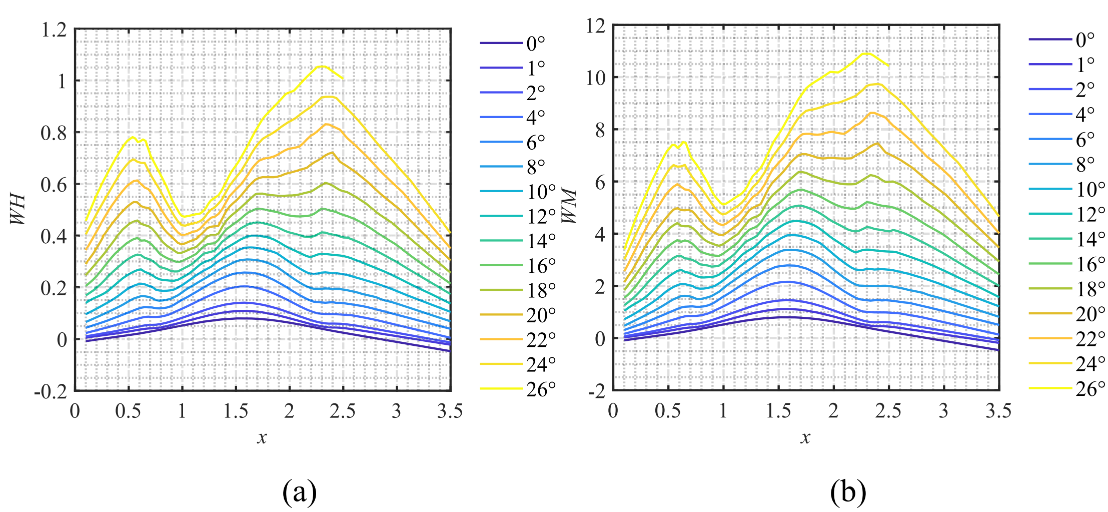

Fig. 3. Complete characteristic curves of the pump-turbine after improved Suter transformation: (a) flow characteristic curve; (b) torque characteristic curve.

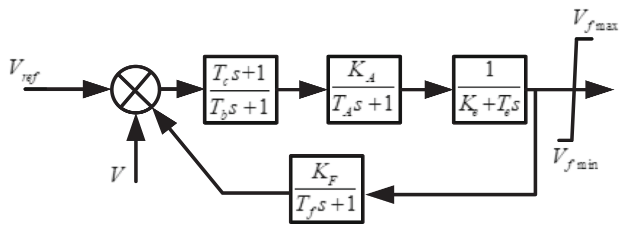

Fig. 4. Excitation system model.

and lag compensation elements, while K_A and T_A denote the gain and time constant of the amplification stage. The variable V_f corresponds to the excitation winding voltage, with K_e and T_e characterizing the exciter's self-excitation properties (gain and time constant respectively). For the rotor stabilization circuit, K_F indicates the gain factor of the soft feedback mechanism, accompanied by its associated time constant T_f.

2.6. Machine-side control model of Full-size converter

The machine-side converter primarily maintains DC-link voltage stability to ensure power balance between the generator and grid sides [24]. It employs a dual-loop voltage-current control strategy, where:

$$\begin{cases} i_d^* = \left(K_p + \frac{K_i}{s}\right)(V_{dc}^* - V_{dc}) \\ i_q^* = \left(K_p + \frac{K_i}{s}\right)(Q_{dc}^* - Q_{dc}) \end{cases} \tag{13}$$

where i_d^* and i_q^* are the reference values of the machine-side current loop; K_p and K_i are the proportional and integral gains of the DC voltage outer loop; V_{dc}^* and V_{dc} are the reference and measured values of the DC voltage; Q_{dc}^* and Q_{dc} are the reference and measured values of the machine-side reactive power.

The current inner-loop control strategy of the machine-side converter employs a combined feedforward compensation and dq-transformation approach, utilizing PI controllers to achieve rapid, error-free stator current regulation. Its mathematical model is expressed as:

$$\begin{cases} u_d = -\left(K_{pi} + \frac{K_{ii}}{s}\right)(i_d^* - i_d) + \omega L i_q + e_d \\ u_q = -\left(K_{pi} + \frac{K_{ii}}{s}\right)(i_q^* - i_q) + \omega L i_d + e_q \end{cases} \tag{14}$$

where u_d and u_q are the dq-axis components of the AC-side voltage of the machine-side converter; K_{pi} and K_{ii} are the proportional and integral gains of the current inner loop; L is the machine-side filter inductance; and e_d and e_q are the dq-axis components of the terminal voltage of the synchronous generator. The control strategy block diagram of the FSC machine side is shown in Fig. 5.

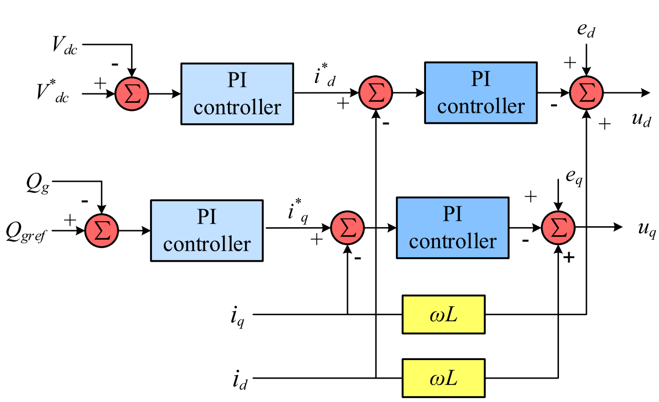

Fig. 5. Block diagram of the machine-side converter control system.

3. Multi-objective optimization of start-up strategies

To promptly respond to grid load variations, PSUs frequently undergo operational mode transitions, particularly during start-up processes. The start-up procedure involves rapid transition from standstill to grid-synchronized steady-state operation, requiring strict adherence to prescribed rotational speed and pressure thresholds. From a control performance perspective, both speed and pressure metrics are critical parameters. However, given the relatively gradual hydraulic pressure changes during start-up, primary emphasis should be placed on achieving both rapid and stable speed acceleration. Furthermore, in shared-penstock heterogeneous-unit configurations, hydraulic interference between units can degrade the otherwise optimal control performance under given operating conditions. Recognizing the inherent trade-off between response speed and stability, this study employs multi-objective optimization to derive a Pareto-optimal solution set for coordinated start-up of heterogeneous units. Decision-makers can then select optimal solutions based on operational priorities. To determine the optimal SUS for parallel-connected and independently-operated units, the rising time of the rotational speed for both variable and fixed units, as well as the ITAE, are selected as objective functions. The unit start-up strategies considered include different start-up strategies, the same SUS with identical parameters, and the same SUS with varying parameters. A multi-objective optimization scheme for the SUS of parallel-connected and independently-operated units is proposed, with the performance indicators of the units serving as constraint conditions.

3.1. Start-up control optimization objectives for PSUs

During the start-up transient process, both the rapidity and stability of unit rotational speed are critical focus parameters. In the open-loop phase, the guide vane opening speed and degree fundamentally determine the rapidity of speed acceleration. During the closed-loop phase, the controller effectively suppresses speed oscillations, thereby ensuring stable grid synchronization. Consequently, this study selects the following dual optimization objectives: (1) the sum of speed rise times for both units, and (2) the aggregate ITAE of rotational speed deviations for both units, formulated as:

$$\begin{cases} \min F_1 = t_{r1} + t_{r2} \\ \min F_2 = \mathrm{ITAE}_1 + \mathrm{ITAE}_2 \end{cases} \tag{15}$$

In the formulation, t_{r1} and t_{r2}represent the speed rise times of the fixed-speed unit and the FSC-VSU, respectively, while ITAE1 and ITAE2 denote the ITAE for the fixed-speed and variable-speed units, respectively.

The first objective function is defined as the sum of the speed rise times of both the fixed-speed unit and the FSC-VSPSU, aiming to minimize the overall start-up duration. The second objective function represents the combined ITAE for both units, ensuring optimal tracking performance during synchronization. Minimizing these dual objectives facilitates smooth grid integration of the co-located units.

3.2. Decision variables for PSU start-up control optimization

In the optimization study of unit start-up strategies, decision variables serve as the key optimization parameters that define and regulate the start-up control logic. This research focuses on analyzing and optimizing two representative start-up strategies: (1) single-stage direct guide vane opening with PID control, and (2) two-stage direct guide vane opening with PID control. Accounting for the operational asymmetry between heterogeneous units during co-start-up in shared-penstock systems, six distinct start-up strategies are systematically

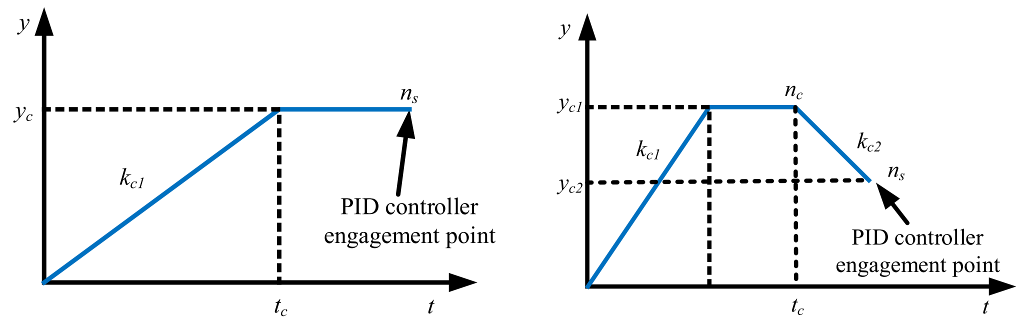

Fig. 6. Schematic diagram of the control strategy for the start-up process of the same pipe and different machines.

designed for the one-penstock-two-unit configuration, as schematically illustrated in Fig. 6.

(1) In the single-stage direct guide vane opening with identical PID parameters control strategy, the guide vanes are opened at the maximum rate k_{c1} to rapidly accelerate the unit speed. Upon reaching the maximum opening position y_c the guide vane opening is maintained constant. As the unit speed increases, the PID controller is engaged when the speed attains the switching threshold n_s regulating and stabilizing the speed at the setpoint. Notably, the speed setpoints differ between the variable-speed and fixed-speed units: the fixed-speed unit targets rated speed, while the variable-speed unit operates at a non-rated speed. Given that k_{c1} is a predetermined parameter, this strategy involves six decision variables: (1) maximum start-up opening y_c, (2) PID controller engagement speed n_s, (3) variable-speed unit setpoint n_{ref}, and (4)–(6) PID control parameters K_p, K_i, and K_d. These decision variables can be formally expressed as:

$$X_1 = \left(y_c, n_s, n_{\mathrm{ref}}, K_p, K_i, K_d\right) \tag{16}$$

(2) In the differentiated single-stage direct guide vane opening with distinct PID parameters control strategy, both units' guide vanes are opened at the maximum rate k_c to achieve rapid speed acceleration. The fixed-speed unit's guide vanes are maintained at maximum opening y_{c1}, while the variable-speed unit's are held at y_{c2}. As the units accelerate, their respective PID controllers engage when the fixed-speed unit reaches switching speed n_{s1} and the variable-speed unit attains n_{s2}, stabilizing each unit at its designated setpoint (with the variable-speed unit targeting a non-rated speed). Consequently, this strategy involves eleven decision variables: (1)–(2) maximum openings y_{c1} and y_{c2}; (3)–(4) engagement speeds n_{s1} and n_{s2}; (5) variable-speed setpoint n_{ref}, and (6)–(11) PID parameters for both units ($K_{\mathrm{p}1}$, K_{i1}, $K_{\mathrm{d}1}$, K_{p2}, K_{i2}, K_{d2}). The decision vector is formally defined as:

$$X_2 = \left(y_{c1}, y_{c2}, n_{s1}, n_{s2}, n_{\mathrm{ref}}, K_{p1}, K_{i1}, K_{d1}, K_{p2}, K_{i2}, K_{d2}\right) \tag{17}$$

(3) In the identical two-stage direct guide vane opening with uniform PID parameters control strategy, the guide vanes are initially opened at the maximum rate k_{c1} to reach intermediate opening yc1 and held constant. This first stage ensures rapid speed acceleration while maintaining stable speed rise characteristics during the initial start-up phase. When the rotational speed attains preset threshold n_c, the guide vanes close at maximum rate k_{c2} to reduced opening y_{c2}, primarily to prevent speed overshoot and instability caused by excessive acceleration. The control mode switches to closed-loop operation upon reaching engagement speed n_s, where the PID controller activates to precisely regulate the unit speed for grid synchronization requirements. Given that k_{c1} and k_{c2} are typically predetermined parameters, this strategy involves eight decision variables: (1)–(2) staged openings y_{c1} and y_{c2}, (3) speed threshold n_c for closing transition, (4) PID engagement speed n_s, (5) variable-speed unit reference n_{ref}, and (6)–(8) PID parameters K_p, K_i, and K_d. These decision variables can be formally expressed as:

$$X_3 = \left(y_{c1}, y_{c2}, n_c, n_s, n_{\mathrm{ref}}, K_p, K_i, K_d\right) \tag{18}$$

(4) In the control strategy of differentiated two-stage direct guide vane opening with distinct PID parameters, the fixed-speed unit opens its guide vanes to the opening degree y_{c11} at the maximum rate k_{c1}. Simultaneously, the variable-speed unit opens its guide vanes to the opening degree y_{c12} at the same maximum rate k_{c1}. Both units then maintain these guide vane positions in subsequent operation. When the fixed-speed unit's rotational speed reaches the setpoint n_{c1}, its guide vanes close to the opening degree y_{c21} at the maximum closing rate k_{c2}. Similarly, when the variable-speed unit's rotational speed reaches the setpoint n_{c2}, its guide vanes close to the opening degree y_{c22} at the same maximum closing rate k_{c2}. Upon the fixed-speed and variable-speed units reaching their respective cut-in speeds n_{s1} and n_{s2}, the control switches to PID regulation. The parameters k_{c1} and k_{c2} are typically known. Consequently, the decision variables include the set openings y_{c11}, y_{c12}, y_{c21}, and y_{c22}; the rotational speed setpoints n_{c1} and n_{c2}; the PID controller cut-in speeds n_{s1} and n_{s2}; the reference rotational speed n_{ref} for the variable-speed unit; and the PID control parameters for both units (K_{p1}, K_{i1}, K_{d1} for the fixed-speed unit and K_{p2}, K_{i2}, K_{d2} for the variable-speed unit), totaling 15 decision variables. The decision variables can be described as follows:

$$X_4 = \left(y_{c11}, y_{c12}, y_{c21}, y_{c22}, n_{c1}, n_{c2}, n_{s1}, n_{s2}, n_{\mathrm{ref}}, K_{p1}, K_{i1}, K_{d1}, K_{p2}, K_{i2}, K_{d2}\right) \tag{19}$$

(5) Under the control strategy combining single-stage and two-stage direct guide vane opening, the fixed-speed unit opens its guide vanes at maximum rate k_c until reaching maximum opening y_{c1}, then maintains this position. PID control activates when the unit's speed reaches cut-in speed n_{s1}, stabilizing at the target speed. The variable-speed unit opens to y_{c12} at rate k_c and holds. When its speed reaches setpoint n_c, the guide vanes close to y_{c22} at maximum rate k_{c2}. PID control engages upon reaching cut-in speed n_{s2}, with the variable-speed unit's reference speed n_{ref} set below rated speed. The 13 decision variables include: guide vane openings (y_{c1}, y_{c12}, y_{c22}), speed setpoint n_c, cut-in speeds (n_{s1}, n_{s2}), reference speed n_{ref}, and PID parameters (K_{p1}, K_{i1}, and K_{d1} for fixed-speed, K_{p2}, K_{i2}, and K_{d2} for variable-speed units). Parameters k_c, k_{c1}, and k_{c2} are typically predetermined.

$$X_5 = \left(y_{c1}, y_{c12}, y_{c22}, n_c, n_{s1}, n_{s2}, n_{\mathrm{ref}}, K_{p1}, K_{i1}, K_{d1}, K_{p2}, K_{i2}, K_{d2}\right) \tag{20}$$

(6) Under the hybrid control strategy integrating two-stage and single-stage direct guide vane opening, the fixed-speed unit initially opens its guide vanes to position y_{c12} at maximum rate k_c and maintains this opening. When the rotational speed reaches threshold n_c, the vanes close to y_{c22} at maximum rate k_{c2} until reaching cut-in speed n_{s1}, whereupon PID control activates. Conversely, the variable-speed unit opens its vanes to maximum position y_{c1} at rate k_c and holds, engaging PID regulation at cut-in speed n_{s2} to stabilize at sub-rated reference speed n_{ref}. The thirteen decision variables comprise: three guide vane positions (y_{c1}, y_{c12}, y_{c22}), one speed threshold (n_c), two cut-in speeds (n_{s1}, n_{s2}), one reference speed (n_{ref}), and two sets of PID parameters (K_{p1}, K_{i1}, and K_{d1} for fixed-speed, K_{p2}, K_{i2}, and K_{d2} for variable-speed units), with k_c, k_{c1}, and k_{c2} typically specified as known parameters.

$$X_6 = \left(y_{c1}, y_{c12}, y_{c22}, n_c, n_{s1}, n_{s2}, n_{\text{ref}}, K_{p1}, K_{i1}, K_{d1}, K_{p2}, K_{i2}, K_{d2}\right) \tag{21}$$

3.3. Start-up control constraints for PSUs

To ensure the practical applicability of the unit start-up control strategy, actual operational limitations must be converted into corresponding constraint conditions. Therefore, this study considers three types of constraints in the multi-objective optimization process for the same-pipe-different-unit SUS.

(1) Boundary constraints of decision variables.

$$X \in [B_L, B_U] \tag{22}$$

where B_L and B_U represent the lower and upper bounds of the decision variable X, respectively.

(2) The settling time constraint refers to the time duration tp from unit startup initiation until rotational speed reaches steady-state operation, with the requirement that this settling time should not exceed a specified threshold value.

$$t_p < T_1 \tag{23}$$

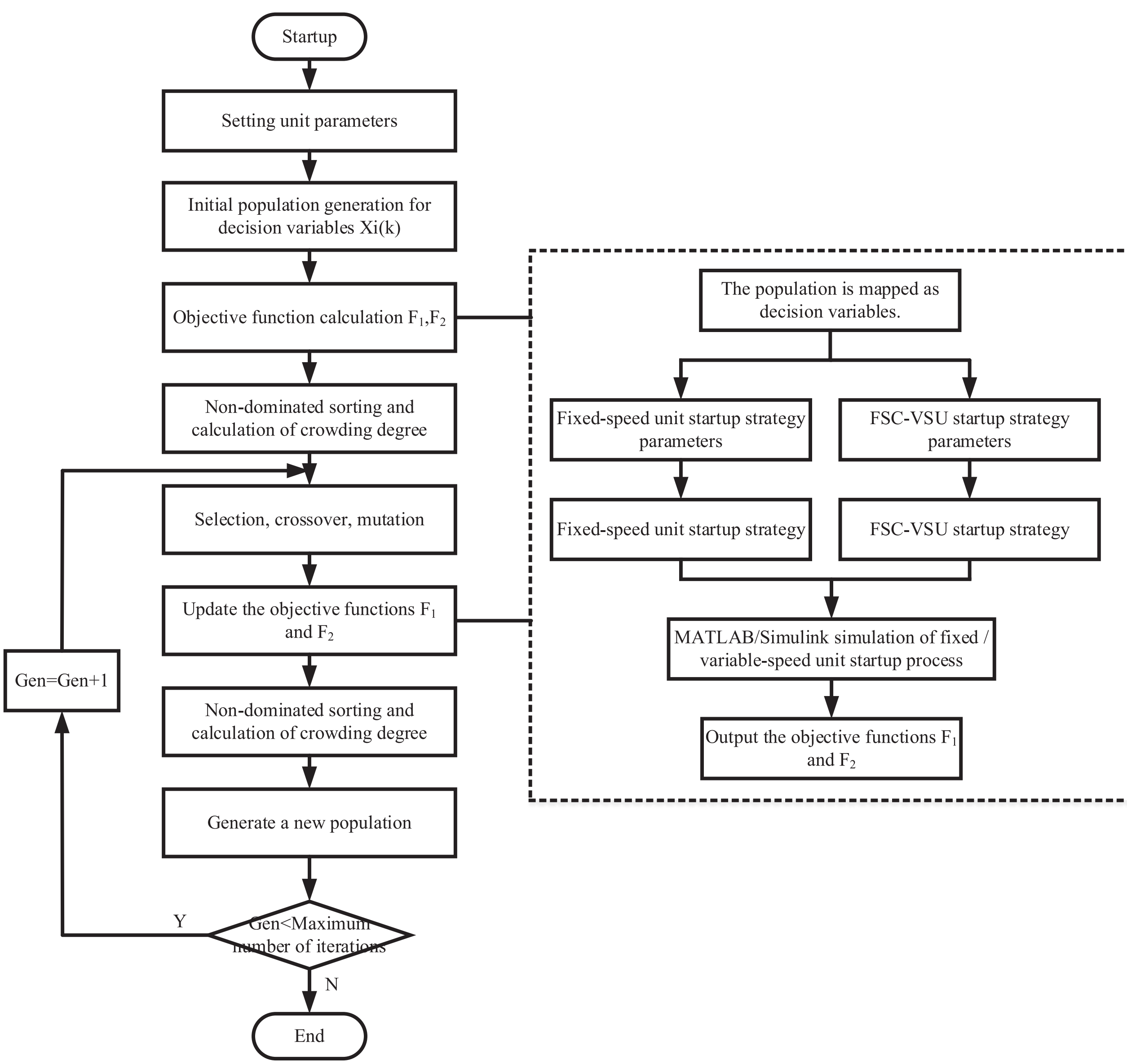

Fig. 7. Multi-objective control scheme optimization for same-pipe and different-machine SUS.

where T_1 denotes the upper limit of the settling time.

(3) The governing guarantee calculation constraint requires compliance with regulatory computational requirements during the transitional process of same-pipe-different-unit startup operations.

3.4. Multi-objective optimization scheme for same-pipeline-different-unit startup strategy

A nonlinear simulation model of the one-pipe-two-unit system was established in MATLAB/Simulink to investigate the same-pipeline-different-unit startup scenario. The proposed control strategy adopts the NSGA-II optimization framework [25], incorporating a penalty mechanism that assigns penalty factors to constraint-violating solutions, thereby ensuring their exclusion from the Pareto optimal set. The complete optimization procedure is illustrated in Fig. 7.

3.5. Decision-making for same-pipeline-different-unit startup strategy

3.5.1. Analytic hierarchy process - Entropy Weight-Game Theory Combined Decision Method

The multi-objective optimization yields a Pareto optimal solution set comprising multiple balanced solutions across competing objectives, providing decision-makers with diversified alternatives for operational selection. To facilitate scientific decision-making, this study employs information entropy theory to quantify and objectively determine indicator weights through metric information analysis, ensuring scientific rigor and objectivity free from subjective bias. Recognizing the value of operational experience, subjective weights are additionally assigned based on empirical knowledge. The combined weighting scheme integrates these objective and subjective weights using game theory to derive optimal composite weights. Finally, the optimal solution is selected by combining the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS).

The entropy weight method [26] is an objective weighting mechanism derived from information entropy theory in information theory. Specifically, this method determines objective weights by analyzing the variability degree of each evaluation indicator. For a given indicator, its information entropy and corresponding weight can be mathematically expressed as follows:

$$\begin{cases} E_j = -\dfrac{1}{\ln n}\sum_{i=1}^{n} p_{ij}\ln p_{ij} \\ w_j = \dfrac{(1-E_j)}{\sum_{i=1}^{n}(1-E_j)} \end{cases} \tag{24}$$

where E_j denotes the entropy of the j-th indicator, p_{ij} represents the proportion of the i-th solution under the j-th indicator, and w_j indicates the weight assigned to the j-th indicator.

Furthermore, game theory is introduced to optimally combine the subjective and objective weights, yielding composite weights. The specific procedure for the game theory-based combined weighting method [27] is as follows:

Step 1: The linear combination of the weight vector $W = [w_1, w_2, \ldots, w_n]$ can be expressed as:

$$W_c = \sum_{i=1}^{n} \alpha_i w_i (\alpha_i > 0) \tag{25}$$

In the formula, α_i is the weighting coefficient, and w_i is the weight determined by different calculation weighting methods.

Step 2: Optimize the weight coefficients α_i of the linear combination to minimize the deviation of W_c from each w_i, that is:

$$\min \left\| \sum_{i=1}^{n} \alpha_i w_i - w_j \right\| (j = 1, 2, \ldots, n) \tag{26}$$

Step 3: Solve Eq. (26) and perform normalization to obtain the final combined weighting coefficients as follows:

$$\alpha_i^* = \frac{\alpha_i}{\sum_{i=1}^{n} \alpha_i} \tag{27}$$

Step 4: The game-theoretic combined weights are derived by applying the final weighting coefficients as follows:

$$W^* = \sum_{i=1}^{n} \alpha_i^* \cdot w_i \tag{28}$$

The optimal solution is subsequently selected by integrating the TOPSIS.

3.5.2. Startup strategy decision model

A hierarchical decision model for developing startup strategies in hybrid variable/fixed-speed PSUs is constructed as illustrated in Fig. 8. This framework is structured hierarchically with three tiers: an objective layer, a criterion layer, and an index layer. The objective layer aims to select the optimal startup strategy. The criterion layer comprises three subsystems, namely the pumped turbine, the pipeline system, and the governor. Accordingly, the indicator layer includes nine indicators: speed overshoot ($\Delta\sigma$), speed rise time (t_r), speed regulation time (t_s), ITAE of rotational speed (ITAE), maximum head fluctuation ($\Delta H_{\max}$), minimum head fluctuation ($\Delta H_{\min}$), water consumption (Q_z), maximum oscillation of guide vanes ($\Delta y_{\max}$), and guide vane travel distance (D_m). Notably, the head fluctuations are produced during closed-loop regulation after PID engagement; water consumption quantifies the flow usage during the startup process; and guide vane travel distance measures the displacement from the initial to the final position [13]. The definitions of speed-related indicators (overshoot time / rise time / settling time / ITAE) are consistent with the traditional definitions of control performance. The computational expressions for water consumption and guide vane travel are derived as follows:

$$\begin{cases} Q_z = \displaystyle\int_{t_0}^{t_3} Q_i(t)\mathrm{d}t \\ D_m = \displaystyle\sum_{i=2}^{n} |y(i) - y(i-1)| \end{cases} \tag{29}$$

Where Q_z represents water consumption, Q_i ($i = 1,2$) denotes the flow rate through the unit, t_0 indicates the initial startup time, t_3 corresponds to the startup completion time, D_m signifies guide vane travel distance, and n stands for the total simulation steps.

4. Case study analysis

To validate the proposed strategy's effectiveness, this study investigates a Chinese pumped-storage power station where both units participate in the startup process: Unit 1 operates as a fixed-speed machine during startup, while Unit 2 functions as a full-size converter variable-speed unit. The simulation replicates the station's operation under no-load conditions with a 560 m working head.

4.1. Optimization parameters

All experiments in this section were conducted under identical head conditions. The multi-objective optimization algorithm parameters were configured as follows: 50 iterations, population size of 30, crossover distribution index of 20, mutation distribution index of 100, and mutation rate of 0.5. Furthermore, the value ranges for the six decision

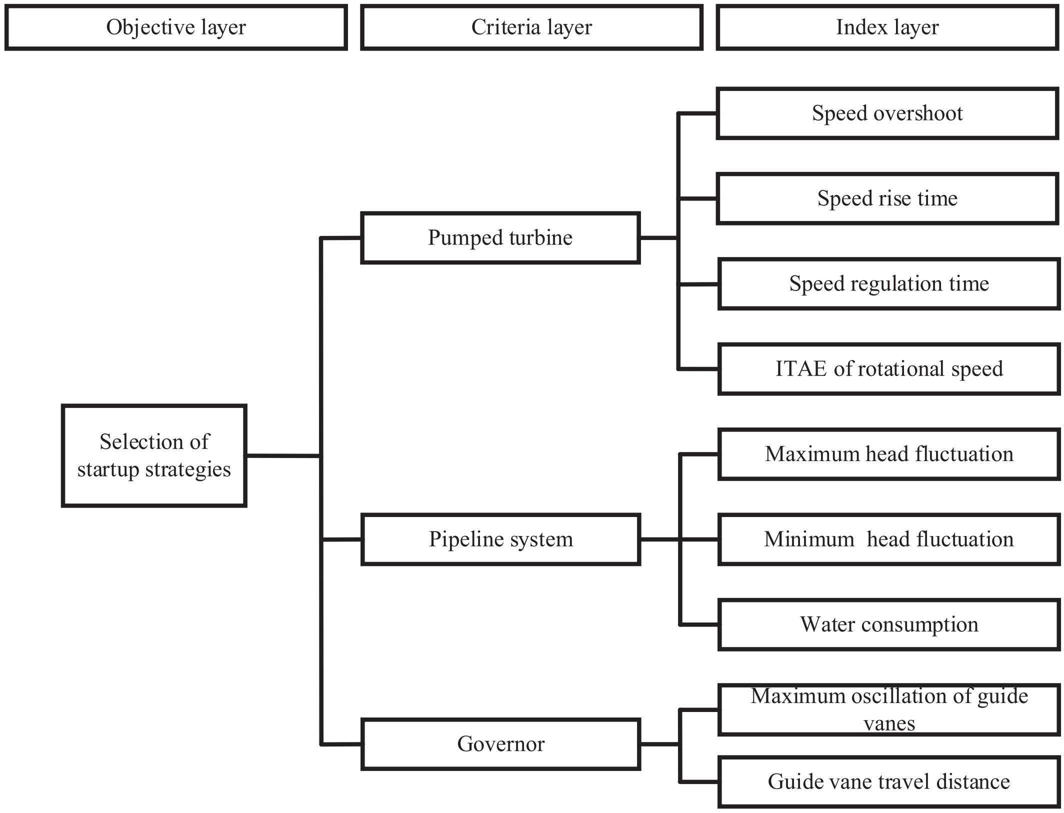

Fig. 8. Decision model of SUS of pumped-storage unit with fixed and variable mixed arrangement.

variables X are specified in Table 1. The setpoint range of the starting speed for the FSC-VSPSU is specified to be within 85% to 100% of the rated speed [28].

4.2. Results analysis

The optimization results yield six Pareto-optimal solution sets corresponding to the proposed startup strategies, as shown in Fig. 9. The analysis reveals an inverse correlation between speed rise time and ITAE, confirming the inherent trade-off between stability and rapidity during startup. Among the six strategies, Strategies 3 and 4 demonstrate superior transient control performance, both employing the two-stage direct guide vane opening with PID control scheme. This indicates enhanced transitional process regulation capability of the two-stage approach. Notably, Strategy 3 produces the narrowest Pareto frontier, attributable to its operational mechanism: larger initial guide vane openings (y_{c1}) accelerate startup under constrained conditions, concentrating solutions near the Pareto frontier's lower bound.

For comparative analysis of the startup strategies' objective performance, this study employs boxplot visualization of the Pareto-optimal solution sets. Fig. 10 demonstrates that Strategy 3 achieves the shortest speed rise time (mean: 24.44 s, interquartile range IQR: 24–26 s), followed by Strategy 1 (mean: 24.65 s). Notably, Strategy 2 exhibits the largest IQR, indicating significant solution dispersion consistent with Fig. 9's distribution. Regarding ITAE minimization, Strategy 4 performs optimally (mean: 4.65, IQR: 2–16). While both Strategies 3 and 4 satisfy transitional requirements, their overlapping Pareto frontiers prevent identification of a dominant solution across both objectives (speed rise time and ITAE), necessitating decision theory for optimal control selection.

To determine the optimal startup control strategy, decision theory is applied to evaluate the Pareto-optimal solution sets of all six strategies. The analytical process involves: (1) calculating objective weights via entropy method, (2) incorporating subjective weights from expert assessment, and (3) deriving composite weights through game-theoretic combination. Table 2 presents the combined weighting values for each strategy's performance indicators, where higher weights indicate greater decision-making importance for corresponding metrics.

Based on the composite weights in Table 2, the TOPSIS was applied to rank the Pareto-optimal solutions. Fig. 11 presents the ranked Pareto-optimal solution sets for the six startup strategies, from which the optimal solutions are identified as Solution 28 (Strategy 1), Solution 20 (Strategy 2), Solution 24 (Strategy 3), Solution 14 (Strategy 4), Solution 18 (Strategy 5), and Solution 17 (Strategy 6). These selections represent neither single-objective optima nor balanced-point compromises, as the proposed decision-making framework holistically evaluates hybrid fixed/variable-speed PSU performance. Comparative analysis shows that: (1) under identical guide vane control laws, customized parameter sets outperform shared parameters; (2) fixed-speed units achieve higher evaluation scores with single-stage rather than two-stage guide vane control. Strategy 4's Solution 14 yields the highest comprehensive evaluation value, thus being identified as the optimal startup strategy.

To validate the superiority of the selected optimal control strategy, a comparative analysis of dynamic characteristics was conducted between the optimal strategy and alternative startup approaches. The top solutions from all six strategies (Strategy 1-Solution 28, Strategy 2-Solution 20, Strategy 3-Solution 24, Strategy 4-Solution 14, Strategy 5-Solution 18, Strategy 6-Solution 17, hereafter referred to as Strategies 1–6) were implemented in the established variable/fixed-speed PSU model. The resulting transient responses - including rotational speed trajectories, startup curves, and excitation voltage transitions - are presented in Fig. 12.

As evidenced by Figs. 12(a) and 12(b), all six startup strategies achieve the target speed within 50s. Fig. 12(a) demonstrates that Strategies 6 and 4 exhibit faster speed acceleration and shorter rise times due to their two-stage guide vane opening pattern, where the initial opening exceeds that of single-stage strategies. Under PID control, speed overshoot persists across all strategies, with Strategies 4 and 5 showing

Table 1
The value range of multi-objective optimization decision variables for startup of PSU with fixed and variable hybrid arrangement.

Startup Strategy1

X_1	y_c	n_s	n_{ref}	K_p	K_i	K_d
BL	0.1	0.6	0.85	0	0	0
BU	0.5	0.9	1	3	3	3

Startup Strategy2

X_2	y_{c1}	y_{c2}	n_{s1}	n_{s2}	n_{ref}	K_{p1}	K_{i1}	K_{d1}	K_{p2}	K_{i2}	K_{d2}
BL	0.1	0.1	0.6	0.6	0.85	0	0	0	0	0	0
BU	0.5	0.5	0.9	0.9	1	3	3	3	3	3	3

Startup Strategy3

X_3	y_{c1}	y_{c2}	n_c	n_s	n_{ref}	K_p	K_i	K_d
BL	0.2	0.1	0.6	0.8	0.85	0	0	0
BU	0.4	0.2	0.8	0.95	1	3	3	3

Startup Strategy4

X_4	y_{c11}	y_{c12}	y_{c21}	y_{c22}	n_{c1}	n_{c2}	n_{s1}	n_{s2}	n_{ref}	K_{p1}	K_{i1}	K_{d1}	K_{p2}	K_{i2}	K_{d2}
BL	0.2	0.1	0.2	0.1	0.6	0.6	0.8	0.8	0.85	0	0	0	0	0	0
BU	0.4	0.2	0.4	0.2	0.8	0.8	0.95	0.95	1	3	3	3	3	3	3

Startup Strategy5

X_5	y_{c1}	y_{c12}	y_{c22}	n_c	n_{s1}	n_{s2}	n_{ref}	K_{p1}	K_{i1}	K_{d1}	K_{p2}	K_{i2}	K_{d2}
BL	0.1	0.2	0.1	0.6	0.6	0.8	0.85	0	0	0	0	0	0
BU	0.5	0.4	0.2	0.9	0.8	0.95	1	3	3	3	3	3	3

Startup Strategy6

X_6	y_{c1}	y_{c12}	y_{c22}	n_c	n_{s1}	n_{s2}	n_{ref}	K_{p1}	K_{i1}	K_{d1}	K_{p2}	K_{i2}	K_{d2}
BL	0.2	0.1	0.1	0.6	0.6	0.8	0.85	0	0	0	0	0	0
BU	0.4	0.2	0.5	0.9	0.8	0.95	1	3	3	3	3	3	3

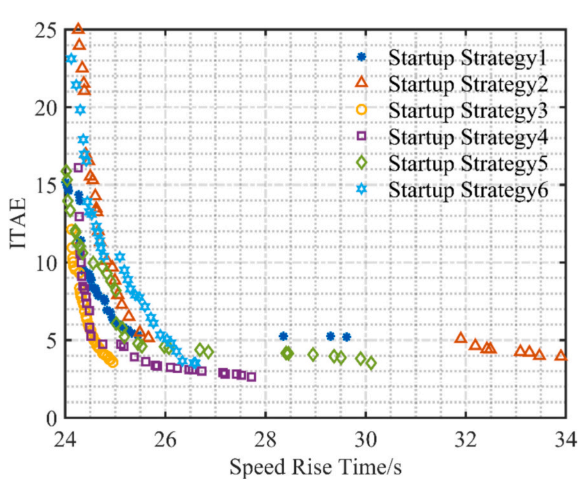

Fig. 9. Pareto optimal solution set for 6 kinds of startup strategies.

minimal overshoot (0.25% and 0.21% respectively). Fig. 12(b) reveals that while the variable-speed unit's reference speed variability precludes direct visual comparison of startup duration, simulation data confirms Strategies 3 and 4 as the fastest (11.43 s and 11.94 s rise times respectively), attributable to their shared two-stage vane control architecture.

Figs. 12(c) and 12(d) reveal distinct operational characteristics between unit types: the fixed-speed unit's trajectory briefly enters but rapidly exits the anti-"S″ region without chaotic behavior, while the full-size converter variable-speed unit's trajectory remains entirely outside the anti-"S″ zone across all strategies. Notably, strategy-dependent reference speed variations yield different stabilization points for the variable-speed unit's trajectory.

Fig. 12(e) demonstrates that the excitation voltage of the full-size converter variable-speed unit stabilizes under all six startup strategies, albeit with strategy-dependent magnitudes due to speed variability. Specifically, the excitation voltage exhibits an inverse relationship with rotational speed – lower speeds yield higher excitation outputs, as evidenced by Strategies 3 and 4 producing the highest voltages corresponding to their comparatively reduced speeds. Despite rotational speed variations, the excitation system maintains tight voltage regulation, with only 0.06 pu maximum steady-state deviation across all strategies.

To better demonstrate the advantages of Strategy 4, Table 3 presents the performance metrics for the same-pipeline-different-unit startup process, including speed overshoot ($\Delta\sigma$), speed rise time (t_r), speed regulation time (t_s), ITAE of rotational speed (ITAE), maximum head fluctuation ($\Delta H_{\max}$), minimum head fluctuation ($\Delta H_{\min}$), water consumption (Q_z), maximum oscillation of guide vanes ($\Delta y_{\max}$),and guide vane travel distance (D_m).

As shown in Table 3, although Strategy 4 exhibits a larger guide vane travel distance compared to Strategies 1, 2, 3, and 5, it achieves the

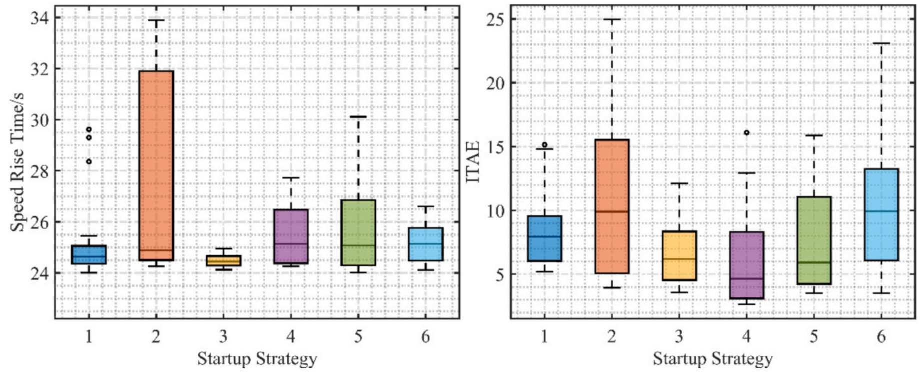

Fig. 10. The box diagram of Pareto optimal solution for 6 kinds of startup strategies.

Table 2
Weight values of six startup policies.

Strategy	Composite weighting of evaluation indicators								
	$\Delta\sigma$	t_r	t_s	ITAE	$\Delta H_{\max}$	$\Delta H_{\min}$	Q_z	$\Delta y_{\max}$	D_m
1	0.17	0.108	0.081	0.085	0.094	0.137	0.154	0.094	0.076
2	0.214	0.087	0.082	0.085	0.096	0.103	0.125	0.105	0.103
3	0.223	0.092	0.096	0.079	0.129	0.085	0.102	0.107	0.087
4	0.195	0.076	0.085	0.085	0.075	0.208	0.115	0.075	0.087
5	0.251	0.093	0.096	0.094	0.097	0.097	0.095	0.089	0.088
6	0.201	0.09	0.087	0.09	0.171	0.123	0.094	0.074	0.071

Note: All weights are normalized; minor deviations in the row sum may occur due to rounding.

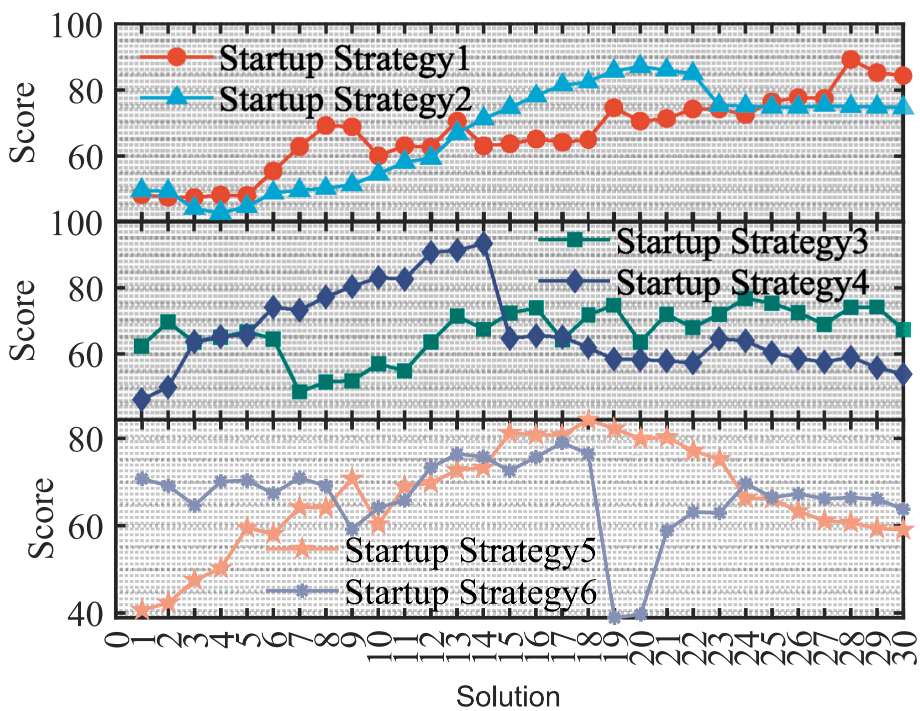

Fig. 11. Scoring results of the Pareto optimal solution sets for six starting strategies.

smallest values in both speed overshoot and minimum head fluctuation indicators. Further comparison reveals that Strategy 4's other performance metrics remain highly competitive with those of the optimal strategies - specifically demonstrating a mere 0.09 s slower speed rise time than Strategy 6 (the optimal in this metric), 0.03 s longer settling time than Strategy 2, 0.26 higher ITAE value than Strategy 3, 12m^3 greater water consumption than Strategy 3, 0.151 larger maximum head fluctuation than Strategy 1, and 0.014 increased maximum guide vane oscillation compared to Strategy 5. These results collectively demonstrate that Strategy 4, despite minor trade-offs in individual parameters, represents the most balanced solution obtained through the decision-making methodology, delivering significantly improved dynamic response quality that enhances both safety and stability during same-pipeline-different-unit startup operations.

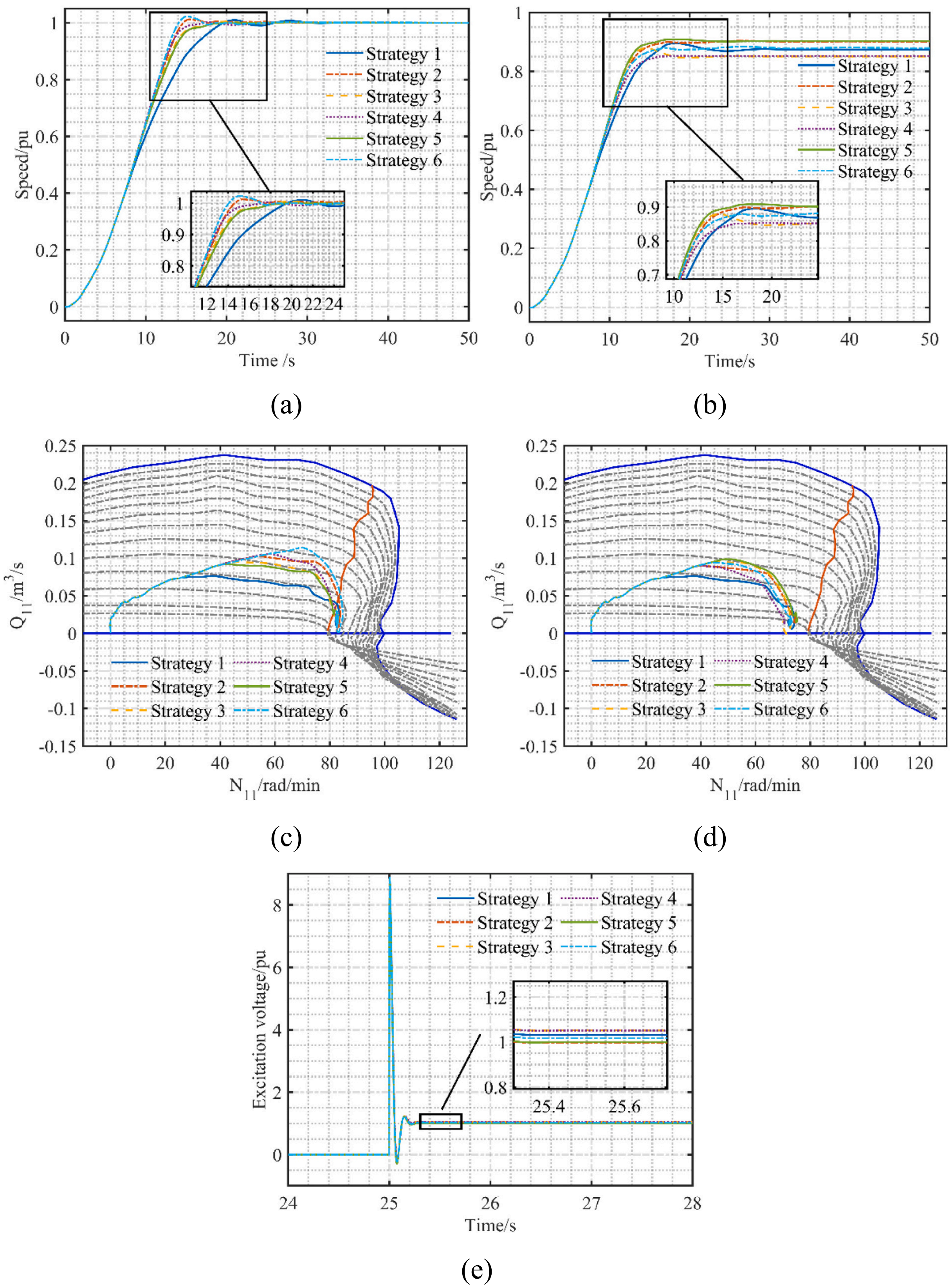

Fig. 12. Transient process curves of key indicators during startup of hybrid ffixed/variable-speed PSUs: (a) speed of the constant speed unit; (b) speed of the variable speed unit; (c) trajectory of unit flow rate for constant speed units; (d) trajectory of unit flow rate of variable speed unit; (e) Excitation voltage of variable speed unit.

Table 3
Performance indicators of the startup process of six startup policies.

Index	Strategy					
	1	2	3	4	5	6
$\Delta\sigma/10^{-2}$	3.13	1.52	3.25	0.35	0.79	2.68
t_r/s	28.36	25.27	24.68	24.75	25.45	24.66
t_s/s	37.86	28.87	31.42	28.9	29.55	30.13
ITAE	5.25	6.5	4.48	4.74	4.77	8.9
Q_z/m^3	1142	1192	1132	1144	1199	1177
D_m/pu	1.023	1.492	1.366	1.601	1.27	1.93
ΔH_{max}/pu	0.127	0.33	0.332	0.278	0.301	0.357
ΔH_{min}/pu	−0.104	−0.073	−0.131	−0.049	−0.054	−0.176
Δy_{max}/pu	0.06	0.072	0.062	0.022	0.008	0.121

5. Conclusions

The increasing integration of intermittent renewable energy has significantly challenged grid load-generation balance, particularly in resolving the stability-speed trade-off during PSU startups. This study establishes a refined hydro-mechanical-electrical coupling nonlinear model for hybrid fixed/variable-speed co-pipeline units with FSC-VSU, capable of characterizing transitional dynamics during startup. A multi-objective optimization framework with decision-making methodology was developed to address the multi-criteria, multi-objective, multi-factor strategy selection problem. Key findings demonstrate that introducing non-rated speed references for variable-speed units effectively avoids anti-S zone operation, resolving low-head speed oscillations, while comparative evaluation shows customized control parameters outperform shared parameters under identical guide vane patterns, with single-stage control superior to two-stage for fixed-speed units, though optimized two-stage schemes prove optimal for co-pipeline operation. The proposed decision model identifies Strategy 4 as superior, exhibiting minimal speed overshoot (0.0035 pu) and head fluctuation (−0.049 pu) alongside dominant performance across other metrics, significantly enhancing startup safety and stability through improved dynamic response quality. The optimization results provide critical decision-making support for formulating startup strategies in same-pipeline-different-unit pumped storage systems, while notably, the proposed multi-objective optimization method exhibits strong universality as a generalized approach applicable to startup strategy optimization for all power plants incorporating variable-speed units.

Nomenclature

t time
H represents the head in the pressure pipeline
Q denotes the flow rate in the pipeline
l the length of the pipeline
A the cross-sectional area of the pipeline
g the gravitational acceleration
f the friction coefficient
D the diameter of the pipeline
a the water hammer wave speed
n_{11}, Q_{11}, M_{11} represent the unit speed, unit flow rate, and unit torque
y the guide vane opening
h, y, n, q, m represent the relative values of head, guide vane opening, rotational speed, flow rate, and torque
T_y the response time constant of the main servomotor
T_a the inertia time constant of the unit
D_f the damping coefficient of the synchronous generator
T_b, T_c the time constants of the lead and lag links
KA, TA the coefficients and time constants of the amplifier
V_f the excitation winding voltage

CRediT authorship contribution statement

Zhang Liu: Writing – original draft, Methodology, Conceptualization. **Diyi Chen:** Writing – review & editing, Supervision, Funding acquisition. **Yunpeng Zhang:** Visualization, Data curation. **Tiantian Wang:** Validation, Software. **Yubi Zeng:** Visualization, Data curation. **Ziwen Zhao:** Visualization, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1
Parameters of units.

Model parameter	Value	Model parameter	Value
Rated head	540 m	Rotational speed	500 r/min
Upstream water level	733–716 m	Downstream water level	181–163 m
Max/Min working head	565/526 m	Rated capacity	306.1 MW
K_1	10	C_h	0.5
K_2	0.8	C_y	0.2

(2) Model validation

To verify the proposed model, an actually operating pumped storage power station in China was selected for a case study, with detailed model parameters available in [29]. The operating conditions for this validation are defined as follows: the upstream water level is 729 m, the downstream water level is 169 m, two units simultaneously execute a 100% load rejection operation, and the guide vane mechanism completes full closure within 40 s. Fig. A1 presents a comparative analysis of on-site measured data and simulated calculation results. The data indicate that: the measured maximum pressure of the spiral case is 860.10 m, while the simulated maximum pressure is 824.19 m, with a relative error of 5.6% between the two; the measured minimum pressure of the draft tube is 14.29 m, and the simulated minimum pressure is 41.88 m, corresponding to a relative error of 4.3%; the measured maximum speed rise is 1.38p.u., and the simulated maximum speed rise is 1.37p.u..The high consistency between the simulation results and on-site measured data directly confirms the effectiveness of the established model. Although minor discrepancies still exist—potentially attributed to measurement errors of the characteristic curves—the consistency in their overall variation trends fully meets the accuracy requirements of engineering practice, which sufficiently demonstrates that the proposed model can accurately reproduce the dynamic response characteristics of the units.

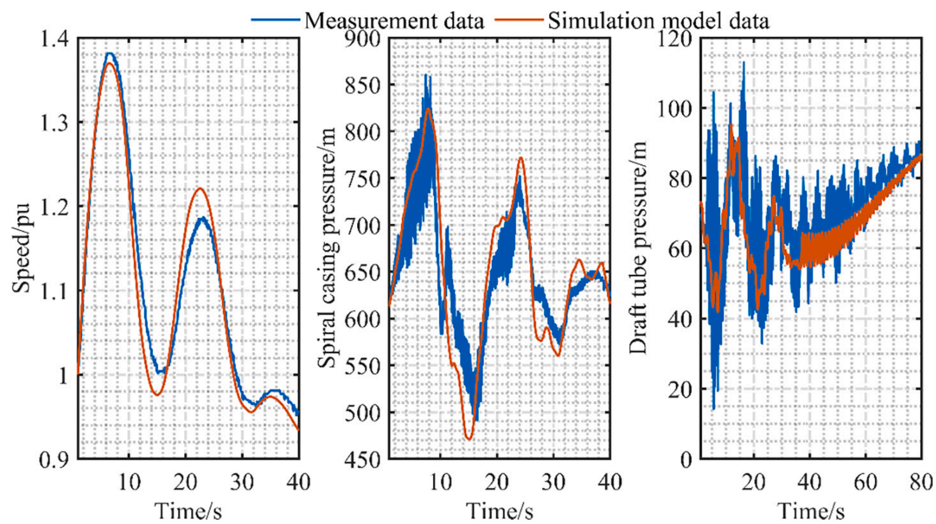

Fig. A1. Validation of the simulation model: (a) speed; (b) spiral casing pressure; (c) draft tube pressure.

Data availability

The data that has been used is confidential.

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